Dense Knowledge Representation: CLIP versus Qwen3-VL Embeddings

Longho Bernard Che | CAP6938, Topic 1

1. Experimental setup

I compare CLIP and Qwen on 24 queries each from four MMEB datasets [1]: CIFAR-100 (classification), CIRR (image retrieval), TextVQA (answer retrieval), and RefCOCO (object-crop retrieval). Seed 6938 controls sampling. Each query has its annotated positive and 31 sampled negatives after text/path deduplication, in shuffled order. Models rank identical pools; these are not official benchmark scores.

`openai/clip-vit-base-patch32` [2] independently encodes images and text; my mixed-input adaptation adds their unit vectors equally and renormalizes. `Qwen/Qwen3-VL-Embedding-2B` [3] jointly processes image/text tokens and normalizes the last non-padding hidden state. Both embed queries and candidates separately, ranking by cosine similarity. Query ablations remove one modality while holding candidate embeddings fixed. The A100 40GB run and full configurations are documented in the notebook.

2. Results

	Accuracy (%)		MRR	
Dataset	CLIP	Qwen	CLIP	Qwen
CIFAR-100	4.2	58.3	.203	.696
CIRR	66.7	87.5	.785	.938
TextVQA	16.7	100.0	.292	1.000
RefCOCO	91.7	100.0	.939	1.000

Full-query accuracy and mean reciprocal rank (MRR) use 24 queries per row; chance accuracy is 3.125%. Overall accuracy is 44.8% for CLIP and 86.5% for Qwen, with 40 CLIP-only failures and 13 shared failures.

Removing the generic classification instruction raises CLIP accuracy from 4.2% to 54.2%, reducing Qwen's advantage to 4.2 percentage points. On CIRR, text-only queries outperform full queries for both models (CLIP: 70.8% versus 66.7%; Qwen: 91.7% versus 87.5%). These pools can reward the edit text without requiring the reference image.

3. Five failure cases

The notebook preserves the inspected images. Below, $F/I/T$ denotes the gold target's rank for full, image-only, and text-only queries; lower is better.

1. CIFAR-100:65 — shared near-tie.

For the gold label *girl*, CLIP predicts *man* ($F/I/T = 2/1/2$) and Qwen predicts *woman* (3/2/4). CLIP's gold-minus-best-negative margin is only -0.0005. Removing the instruction restores rank 1, identifying the added text vector as the cause of the reversal. Qwen also worsens with text. The coarse 32×32 portrait and ambiguous labels limit stronger claims.

2. CIFAR-100:69 — fusion loses recognition.

For pink flowers labeled *tulip*, CLIP predicts the literal dataset label `cra` (18/1/31); Qwen retrieves *tulip* (1/1/13). CLIP's gold margin drops from $+0.0286$ for image alone to -0.0816 after fusion. Its image embedding already recognizes the flower: adding generic text distorts the ranking. Qwen retains the correct match through joint conditioning, although candidate wording remains a confounder.

3. CIRR:193 — incomplete attribute binding.

The reference shows four dogs in a car; the edit requests one dog in a field. CLIP retrieves several dogs on a deck (7/5/15), missing count and setting. Qwen retrieves one dog on a sofa (2/12/1), satisfying only count. Qwen's text-only success disappears when the reference image returns, suggesting competing appearance and edit cues in its pooled representation. Joint encoding does not reliably bind every requested attribute.

4. TextVQA:33 — selecting relevant text.

Asked the time on a smartwatch, CLIP predicts notification text `say thnx` (6/2/12); Qwen retrieves the small, partly glare-obscured `13:57` (1/16/4). Qwen succeeds only with both modalities, consistent with question-conditioned selection of visual text. CLIP's independent image encoder cannot adapt its features to the question. Different resolutions and OCR-related training also contribute plausible explanations.

5. RefCOCO:595 — object versus scene.

For "white chair top right," CLIP retrieves diners (5/5/2), whereas Qwen retrieves the chair crop (1/4/1). CLIP's global scene similarity can outweigh the requested object because its visual features are not word-conditioned. However, Qwen also succeeds from text alone: crop appearance can identify the target. This supports language-to-crop matching without demonstrating spatial reasoning. The metric measures retrieval, not localization IoU.

4. Conclusion

Qwen wins more full-query rankings; TextVQA supports useful joint conditioning, while CLIP's fusion rule causes substantial failures despite useful image embeddings. Small samples, 32-way pools, and training/preprocessing differences prevent causal attribution to architecture. Fusion-weight and harder-negative tests could clarify these effects.

References

[1] Z. Jiang et al. *VLM2Vec: Training Vision-Language Models for Massive Multimodal Embedding Tasks.* 2024. arXiv:2410.05160.

[2] A. Radford et al. *Learning Transferable Visual Models From Natural Language Supervision.* 2021. arXiv:2103.00020.

[3] M. Li et al. *Qwen3-VL-Embedding and Qwen3-VL-Reranker: A Unified Framework for State-of-the-Art Multimodal Retrieval and Ranking.* 2026. arXiv:2601.04720.